

India: A potential market with competitive advantages

There is tremendous opportunity for programmable technology in India, especially with rapidly growing new markets in diverse domains like communication, industrial, medical, defence and military. Gangatharan Gopal, country manager for Altera India, says, "There are new innovative products designed by Indian companies like fuel-dispenser systems used at petrol stations, thermal printer designs used in shopping malls for printing the receipts, LED-based outdoor display boards, railway passenger information displays, patient monitoring systems and many more. Programmable hardware like FPGAs have become heart of the system."

However, Shah feels that India is yet to explore the FPGA market to the same extent as rest of the world. There are customised products that can be designed using FPGAs especially for the Indian market at effectively low cost. "The new products today differ only in terms of looks and some add-on features from their previous version or release. As the product shelf life is shortening, you have to lower your initial investment cost. This kind of market will drive the use of FPGAs as you can get the flexibility of using the same device, re-programming it and providing new solution," suggests Shah. With FPGA usage, you can have low investment cost and low volume catering to the Indian market.

Many vendors like Lattice Semiconductor and Altera are coming up with low-cost FPGAs targeted at the Indian market. With in-built IP cores dedicated to Indian applications, the sales of these low-cost solutions will soon pick up in India.

features on chip, like RAM and digital signal processing (DSP) blocks, memory controller, high-speed serial input/output transceiver and hard intellectual property (IP) for PCIe and Ethernet MAC," informs Tso. "There is an extensive library of IP available to use and as the technology improves, there would be more and more features adding to it," Varma adds.

FPGAs with embedded DSP blocks

Digital signal processing is required in every embedded application starting from audio-video devices to defence equipment. Any digital signal processing algorithm can be broken into the simplest architecture of multiply-and-accumulate function. Integrating these multipliers and accumulators on board with FPGAs serves a high-end alternative for DSP-intensive applications.

The natural question that arises is whether you can replace a DSP processor with an FPGA having a DSP block. The answer is both 'yes' and 'no.' "The difference between a DSP processor and a DSP-implemented FPGA is the same as between a software and a hardware. DSP is still a programmable processor and you write software programs that it runs," says Varma. He adds, "FPGA is a programmable

hardware where everything runs concurrently. If you were to design a 256-tap filter, in DSP you will have to run the filter 256 times as it runs the program sequentially. In FPGA, as it is hardware, the 256-times run happens simultaneously in a single clock cycle." FPGAs are uniquely suitable for repetitive DSP tasks as they operate in parallel. However, dedicated DSPs are sometimes preferred because of their low cost in applications that do not demand repetitive and high performance.

Telecom technology is one of the early adopters of FPGA for high-performance DSP applications. With the versatility of FPGAs, you are able to provide support for multiple platforms leading to a universal handset for different technologies (GSM, CDMA, 3G and 4G). FPGAs are also used in networking devices like wireless base stations, modems, switches and routers.

"After evaluating the pros and cons of DSPs and FPGAs, we found that FPGAs are the most suitable for our multiple-input multiple-output (MIMO) system design. Our latest design of digital baseband for 4x4 MIMO system supports 180 MHz on FPGA with 60 per cent utilisation, which is very good," informs Aggarwal.

With high-definition images and

3-D rolling, and growth in consumer electronics devices like the plasma TVs and digital cameras, there is a need for image processing algorithms that produce high-quality image. Image processing operations are simple and repetitive, so these can be best implemented using FPGAs with DSP block. But in image pipelining where 'blobs' and 'region of interest' in an object are inspected, a DSP may be preferred because of the varying sizes of blobs and subsequent processing required.

There are certain scenarios where you will see a DSP and an FPGA being used together. Here, the algorithms are divided in computing-intensive tasks for FPGAs and others for the processor. This is called 'hardware exploration,' shares Varma.

Xilinx' Virtex-5, Virtex-6, Spartan-6 and Extended Spartan-3A, Actel's RTAX, Altera's Cyclone and Stratix are some of the FPGA families with embedded multiplier of varying bit-length used for digital signal processing applications.

High-speed serial I/O technology

The data throughput requirements and chip-level functionality are increasing as high-speed communications start using FPGAs. This has compelled the industry to migrate from lower-data-rate parallel connections to higher-speed serial connections. This concept of SerDes (serialiser-deserialiser) introduced by Lattice involves transmitting serialised data over high-speed, differential pairs rather than lower-speed parallel buses, facilitating higher data rates with fewer pins.

Shakeel Peera, director of marketing for high-density solutions, Lattice Semiconductors, shares, "There was an increasing need for low-cost FPGAs. However, these did not have high-speed interfaces. Through the concept of SerDes, we were able to provide low-cost FPGAs with high-speed serial interface." SerDes serves a wide market, including wired and wireless communications (like microwave radio